**DATABASE of System**

**Overview**

This Database folder contains the complete data design of the Smart Parking System. It includes the Entity Relationship Diagram (ERD) and database schema that define how data is stored and managed.

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**Database Entities**

| Entity | Description |
|---|---|
| **User** | Stores student and faculty information |
| **Admin** | Manages system operations |
| **ParkingSlot** | Represents available parking spaces |
| **Reservation** | Stores booking details |
| **Notification** | Handles system messages and alerts |

**Relationships:**

| Relationship | Type | Description |
|---|---|---|
| **User → Reservation** | One-to-Many | A user can make multiple reservations |
| **ParkingSlot → Reservation** | One-to-Many | A slot can be reserved multiple times over time |
| **Admin → System** | One-to-Many | Admin manages system records |

The database design follows the principles of normalization to reduce redundancy and improve efficiency. Each table is structured to maintain atomicity, consistency, and referential integrity. For example, user information is stored separately from reservation records, and each reservation is linked to a specific parking slot.

This design ensures efficient data retrieval, accurate slot tracking, and reliable system performance. The database acts as the backbone of the system, supporting all operations such as booking, availability updates, and administrative reporting.

**Design Principles:**

| Principle | Description |
|---|---|
| **Normalization** | Eliminates redundant data |
| **Referential Integrity** | Maintains valid relationships between tables |
| **Data Consistency** | Ensures accurate and updated information |
| **Atomic Transactions** | Ensures safe booking operations |